RSS DAIRY COW MONITORING SYSTEM

A Portable Backpack RFID-Based Smart Dairy Farm Monitoring Solution | DETD Project Summary

<table>
<tr><td>Project Title</td><td>RSS Dairy Cow Monitoring System</td></tr>
<tr><td>Students</td><td><table><tr><td>Gagan Singh</td><td>2301372</td></tr><tr><td>Apar Saran</td><td>2301362</td></tr></table></td></tr>
<tr><td>Aims & Objectives</td><td>To develop a portable, backpack-worn RFID system that enables dairy farm workers to monitor their herd in real time — without any fixed infrastructure — by combining RFID tag identification with an RFID reader and a mobile application.
Core Objectives:
Cow Entry/Exit Counting — Scan RFID ear tags at gates using the backpack reader to auto-count cows entering or leaving the dairy.
Zone-Based Counting — Maintain real-time occupancy counts for the Bada (enclosure) and grazing fields as workers walk through each zone.
Live Location Tracking — Track approximate real-time locations of cows inside the Bada via RFID scan history displayed on the mobile app.
Health & Safety Monitoring — Identify potentially sick or distressed cows through scan gap detection and zone inactivity alerts on the mobile app.</td></tr>
<tr><td>Scope</td><td>What was done in this project:
Designed a wearable backpack RFID system consisting of an RFID reader unit and antenna carried by the farm worker, paired wirelessly with a mobile application.
Each cow is fitted with a passive RFID ear tag for unique identification.
The mobile app receives scan data from the backpack reader, maintains per-zone cow counts (Bada / Grazing Field), and displays cow location history.
All data is stored locally on the device — the system operates fully offline with no internet or cloud dependency.
Workers can manually upload or reconcile scan records across shifts.
Health alerts are generated based on anomalies in scan frequency and zone presence data.</td></tr>
<tr><td>Applications</td><td>Dairy farms seeking affordable, infrastructure-free livestock monitoring.
Rural farm environments where internet connectivity is unavailable or unreliable.
Multi-zone farms requiring automated headcounts across grazing fields and enclosures.
Animal welfare programs requiring early detection of sick or isolated animals.
Scalable to large herds by deploying additional RFID ear tags with no hardware changes.</td></tr>
<tr><td>DETD Alignment</td><td>This project embodies the Design Engineering / Theme Development (DETD) philosophy in the following ways:
Problem-Driven Design: The system is built around a real, observed pain point — manual and error-prone herd management in Indian dairy farms — and engineers a hardware-software solution to address it directly.
Portability as a Design Constraint: Fixed infrastructure was explicitly rejected as a design choice, pushing the team to engineer a wearable backpack RFID system — a non-trivial hardware design decision that balances antenna range, reader power, and worker ergonomics.
Systems Integration: The project integrates hardware (RFID tags + backpack reader), communication (RFID protocol), and software (mobile app with local storage and alert logic) into a coherent, end-to-end solution.
Real-World Applicability: The system targets rural dairy farms across India where cost, connectivity, and ease of deployment are critical — directly aligning with the DETD theme of engineering for impact and accessibility.</td></tr>
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